Praveen Kumar

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Location: Hyderabad

Professional Summary

Highly motivated Machine Vision and Embedded AI engineer with experience in developing real-time inspection and monitoring systems, seeking a Junior Computer Vision Engineer role at AegisVision. Proficient in deploying computer vision models on edge devices and integrating thermal, ultrasonic, and optical sensors for defect detection and predictive maintenance systems. Excited to leverage my skills in YOLOv8, OpenCV, PyTorch, and TensorFlow to contribute to AegisVision's innovative projects.

Technical Skills

- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Keras
- **Computer Vision:** YOLOv8, OpenCV, Object Detection, Image Processing
- **Edge AI:** Raspberry Pi, Embedded Inference, Real-Time Processing
- **IoT & Embedded:** Arduino, Sensor Integration, Control Systems
- **Sensors:** LiDAR, Ultrasonic, Thermal, Optical, Gyroscope
- **Tools:** Git, Jupyter Notebook, Twilio API, CAD
- **Programming:** Python, Java, SQL

Education

Bachelor of Technology in Computer Science (AI & ML), Woxsen University, Hyderabad, 2022 – 2026

Key Projects

Railway AI Research (Patent Published): Wheel Condition Assessment and Maintenance Device

- Designed AI-enabled wheel inspection system using machine vision and sensor fusion
- Integrated LiDAR, thermal, ultrasonic and optical sensors for defect detection
- Developed microcontroller-based real-time diagnostic and IoT monitoring system
- Implemented predictive maintenance approach using sensor fusion data

Smart Crop Guard – Real-Time Edge AI Monitoring System

- Developed embedded machine vision system using YOLOv8 achieving sub-200 ms inference latency

- Built full AI pipeline including dataset preparation, training and edge deployment
- Integrated sensors, lights and speakers with automated alerts via Twilio API
- Designed temporal filtering logic reducing false detections

AI-Powered Smart Irrigation System

- Developed EfficientNet-B0 model achieving 90%+ accuracy for crop-stage classification
- Built real-time inference pipeline on Raspberry Pi with camera and soil sensors
- Designed offline irrigation decision logic using vision and sensor data

Achievements & Publications

- **IEEE Publication** – Smart Crop Guard: AI-Enabled Wildlife and Adaptive Deterrence System
- **Patent Granted** – Sun Tracking Solar Panel System
- **Patent Published** – Wheel Condition Assessment Device (Application No:-202541067091)
- **Co-author** – IoT Applications Book Chapter Role of IoT In Communications (Taylor & Francis)
- **Treasurer** – IEEE Student Branch
- **Shortlisted** – MSME Zonal SIH 2024